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1 Smart Attendance Checker

```
import Java.util.Scanner;
public Class Main{
    public Static void main (String [] ){
    Scanner SC = new Scanner (System.in);
    System.out.print ("Enter total classes conducted: ");
    int total classes = SC. nextInt();
    System.out.print ("Enter classes attended: ");
    int attended Classes = Sc.nextInt ();
    double attendance percentage =((double) attendedclass / total
classes ) * 100;
    System.out.println (" Attendance Percentage: " + attenda
nce percentage + " %.");
   if (attendancepercentage >= 75) {
     System.out.println ("Eligible for Exam");
  } else {
     System.out.println ("Not Eligible");
  }
  Sc.close();
 }
}
```

Output

```
Enter total Classes Conducted: 88
Enter Classes attended: 75
Attendance Percentage: 85.22723%
Eligible for exam
```

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2. AI age predictor

```
import Java.Util.Scanner;
Public class Main {
    public Static void main (String [] ){
        Scanner SC = new Scanner (System.in);
        System.out.print ("Enter your current age: ");
        int age = SC.nextInt ();
        System.out.Println ("\n--- AI Age predictor ---");
        System.out.println ("your age after 10 years: " +
        (age + 10));
        System.out.println (" your age after 25 years: " + (age + 25));
        System.out.Println (" your age after 50 years: " + (age + 50));
        int currentYear = 2026;
        Int yearTurn100 = currentyear + (100 - age);
        System.out.println ("You will turn 100 in the year: " +
year Turn100);
        SC.close ();
    }
}
```

Output

```
Enter your current age: 21
--- AI Age predictor ---
your age after 10 year: 31
your age " 25 year: 46
your age " 50 year: 71
you will turn 100 in the year: 2105
```

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3: fitness Challange tracker

```
import Java.util.Scanner;
Public class Main{
  public static void main (String [] ){
  Scanner sc = new Scanner (System.in);
  int totalSteps = 0;
  int highestSteps = 0;
  for (int day = 1; day <= 7; day ++) {
    System.out.print ("Enter Steps walked on Day" +day +": ");
    int steps = Sc.nextInt();
    total steps += steps;
    if (Steps > highestSteps){
      highestSteps = steps;
    }
  }
  double averageSteps = total Steps / 7.0;
  System.out.println ("\n ---fitness Challenge Report ----");
System.out.println ("Total Steps: "+ totalSteps);
System.out.println (" Average Steps: "+ averageSteps);
System.out.println (" Highest steps in a Day: " + highest steps);
sc.close ();
}
}
```

Output

Enter steps walked on Day 1 : 1000
" " " " " 2 : 2000
" " " " " 3 : 3000
" " " " " 4 : 4000
" " " " " 5 : 5000
" " " " " 6 : 6000
" " " " " 7 : 7000

--- Fitness Challange Report ---
Total Steps: 28000
Average steps : 4000.0
Highest Steps in a Day : 7000

4 Fibonacci Series

```
impat Java.Util.Scanner;
public class Main{
  public Static void main (String [] ){
    Scanner SC = new Scanner (System.in);
   System.out.print("Enter the value of n: ");
   int n = SC.nextInt();
   int first = 0; second = 1;
  System.out.println("fibonacci Series:");
  For (int i = 1; i <= n; i++){
   System.out.print(first +" ");
   int next = first + second;
    first = second;
    second = next;
  }
 SC.close();
  }
}
```

Output

```
Enter the value of n: 10
Fibonacci Series:
0 1 1 2 3 5 8 13 21 34
```

5 Prime number check 6

```
import java.util.Scanner;
public class Main{
    public static void main (String [ ]) {
    Scanner.SC = new Scanner(system.in);
   System.out.print ("Enter a number: ");
   int num= SC.next Int();
   boolean isprime =true;
    if(num <=1){
      isprime =false;
     }else {
     for (int =i =2; i<= num /2;  i++){
      if (num %i ==0){
       isprime = false;
        break;
      }
    }
  }
  if(is prime){
  System.out.println(num +" is a prime number.");
  }else {
  System.out.println (num +" is not a prime number.");
 }
 sc.close();
 }
}
```

Out put
Enter a number: 1
1 is not a prime number.

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6 logical operaters check

```
impurt java. util. Scanner;
Public class Main{
  public static vaid main (String [ ]){
  Scanner SC=new Scanner (System.in);
   int num1 = SC.nextInt ();
  System.out.print("Enter Second number: ");
  int num2 = SC. nextInt();
  System.out.print("Enter third number: ");
  int num3 = SC.nextInt();
  boolean result = (num1 + num2 = =num3);
  if (result){
   System.out.println("The third number is the sum of the first
two number: ");
}else{
 System.out.println("The thirdnumber is not the sum of the
first two number: ");
}
sc.close ();
 }
}
```

output

```
Enter first number:2
Enter Second number:2
Enter third number:4
The third number is the sum of the first two
numbers.
```

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7. Multiplication Table

```
import Java. Util. Scanner;
public class main {
  Public static void main (String [] ){
    Scanner sc= new Scanner (System.in);
    System. out. print ("Enter a number : ");
    int num = sc. nextInt ();
   System.out.println ("\n Multiplication Table of" + num);
   for (int i = 1; i <= 10; i++){
     System. out.println (num +" x " +i+" = "+ (num *i));
   }
   sc.close ();
    }
  }
```

Multiplication Table of 2

2x1 = 2
2x 2 = 4
2x 3 = 6
2x 4 = 8
2x 5 = 10
2x 6 = 12
2x 7 = 14
2x 8 = 16
2x 9 = 18
2x 10 = 20

8. Right - Angle Triangle Pattern

```
import java.util.Scanner;
public class main{
    public static void main (String [] ){
    Scanner sc = new Scanner (system.in);
    System.out.print("Enter the number of rows: ");
    int rows = sc.nextInt();
    for (int i =1; i <= rows; i++){
     for (int j =1; j <= i; j++){
      System.out.print ("*");
     }
     System.out.print ();
    }
    sc.close ();
  }
}
```

output

```
Enter the number of rows: 5

*
**
***
****
*****
```

5. Pyramid Pattern

```
import java.util.Scanner;
public class Main{
    public static void main (String [ ] ){
    Scanner sc = new Scanner (System.in);
    System.out.print ("Enter the number of rows: ");
    int rows = sc.nextInt();
    for (int i = 1; i <= rows; i++){
      for (int j = 1; j <= rows - i; j++){
        System.out.print (" ");
      }
     for (int k = 1; k <= (2 * i - 1); k++){
       System.out.print ("*");
     }
     System.out.println();
    }
   sc.close();
  }
}
```

Output

```
Enter the number of rows: 5

    *
   * * *
  * * * * *
 * * * * * * *
* * * * * * * * *
```

10. Palindrome Number

```
import Java. util. Scanner;
public class Main{
  public static boolean ~~is p~~ ispalindrome(int x){
    int original = x;
    int reversed = 0;
    while (x>0){
      int digit = x % 10;
      reversed = reversed * 10 + digit;
      x = x/10;
    }
    return original == reversed;
  }
  public static void main (String []){
  Scanner sc = new Scanner (System.in);
  int num = sc. nextInt();
  i~~f num = sc. nextInt~~
  if( ispalindrome (num)) {
    System.out.println(num+" is a palindrome Number.");
  }else{
    System.out. println (num +" is not a palindrome Number.");
  }
  sc.close();
  }
}
```

Output

Enter a number: 1

1 is a palindrome number.

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11. Basic Calculator

```
import java.util.Scanner;
public class Main {
  public static void main (String [] ) {
  Scanner sc = new Scanner (System.in);
  System.out.print ("Enter first number : ");
  double. num1 = sc.nextDouble ();
  System.out.print ("Enter Second number : ");
  double num2 = sc. nextDouble ();
  System.out.print ("Enter operator (+, -, *, /): ");
  char operator = sc. next().charAt (0);
  double result;
  switch (operator) {
     case '+' :
        result = num1 + num2;
   System.out.println ("Result = " + result);
   break;
   case '-';
   result = num1 - num2;
  System.out. println ("Result = " +result);
  break;
  case '*' :
  result = num1 * num2;
System.out. println (" Result =" + result);
  break
  case "/';
  if (num2 ! =0) {
      result = num1 / num2;
```

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```
System.out.println("Result = " + result);
} else {
  System.out.println("Error! Division by zero is not
allowed.");
}
break;
  default:
  System.out.println("Invalid operator!");
}
sc.close();
}
}
```

output

```
Enter first number: 1
Enter second number: 2
Enter operator (+, -, *, /): +
Result = 3.0
```

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12. Grade Calculator

```
import Java. util. Scanner;
public class Main{
  public static void main (String [] ){
  Scanner sc = new Scanner (System.in.);
  System. out. print ("Enter marks: ");
  int marks = sc. next Int ();
  String grade =
       (marks >= 90) ? "A+":
       (marks >= 85) ? "A":
       (marks >= 80) ? "A-":
       (marks >= 75) ? "B+":
       (marks >= 70) ? "B":
       (marks >= 65) ? "B-":
       (marks >= 60) ? "C+":
       (marks >= 55) ? "C":
       (marks >= 50) ? "C-":
       (marks >= 45) ? "D+":
       (marks >= 40) ? "D":
         "F";
  System. out. println ("Grade : " + grade);
  sc.close ();
  }
}
```

Output
Enter marks: 85
Grade: A